Singular: Multi-Timescale Autoregressive Generation

Subject Area: Deep Learning Architectures, Cognitive Scaling Laws, and Non-Linear Dynamical Systems

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ABSTRACT

Modern Autoregressive Large Language Models (LLMs) operate under a uniform temporal scale, forcing every generated token—regardless of its informational entropy—to evaluate synchronously across the entire dense parametric backbone. This monolithic, single-rate sequence processing subjects high-order semantic representations of the network to low-level grammatical, syntactic, and structural token constraints. This uniform temporal flattening potentially leads to structural synaptic capacity redundancies and may induce a fundamental trade-off between open-world alignment and emergent generative sequence entropy (the so-called "alignment tax"). Drawing on non-linear singular perturbation theory, this paper formalizes **Multi-Timescale Autoregressive Generation**—a coupled dual-timescale architecture instantiated as the **Singular State-Space Model (Singular-SSM)** [1, 2]. We model high-order logical planning as a discrete-time slow semantic invariant manifold ($\sim 2\,\text{Hz}$), while routing continuous-time fast-rate token transitions through a selective state-space boundary-layer core ($\sim 50\,\text{Hz}$). We establish a cross-timescale Selective Resonance parameter injection mechanism driven by continuous-time Predictive Error Resonance Gating, where the slow clock step-size is controlled by the fast layer's integrated prediction error. Our theoretical analysis and simulations suggest that isolating high-entropy semantic abstractions from routine low-level execution can reduce the effective degrees of freedom within the primary latent space, thereby providing a control-theoretic substrate for future investigation of alignment tax mitigation and System 2 deliberation in multi-timescale state spaces. **Not every token needs the full model.**

I. INTRODUCTION & THE COGNITIVE LOCK-IN OF SINGLE-RATE SSMS

Contemporary sequence models—including both Transformers and modern Selective State-Space Models (SSMs) like Mamba—dominate autoregressive prediction via a flattened, uniform temporal structure [1, 2, 3]. Whether executing a multi-step out-of-distribution (OOD) logical deduction or generating a predictable grammatical connector (e.g., punctuation marks or high-frequency functional phrases), the complete deep parameter stack (W_{total}) is executed uniformly per token step. This monolithic processing graph may introduce a structural *entropy-weight mismatch*: the parameter footprint of the network can be heavily dominated by local syntactic constraints, micro-spelling codes, and surface-level textual generation tasks.

Consequently, optimizing next-token accuracy under rigid single-rate execution forces a substantial fraction of the dense parameter space into modeling low-level token dependencies, constraining the latent capacity available for long-range, cross-domain semantic synthesis.

Furthermore, this synchronous temporal design is hypothesized to underpin the contemporary "Alignment Tax." To guarantee safety and factual alignment, engineering interventions such as Reinforcement Learning from Human Feedback (RLHF) [4] and Direct Preference Optimization (DPO) [5] contract the variance of the global autoregressive phase space. Because the network's parameters are shared uniformly across all timescale task layers, this structural contraction of the latent space can restrict the model's underlying open-world exploratory capacity and original logical flexibility, leading to a reduction in generative sequence entropy.

To address this scale lock-in, it may be beneficial to move beyond local attention window variants or engineering pruning, which leave the underlying uniform temporal clock unaddressed. We hypothesize that efficient cognitive processing may be facilitated not by a heavier single-clock operator, but by the mathematical decoupling of the processing spectrum into heterogeneous, asynchronously coupled timescales, supporting the emergence of a dual-system cognitive structure [6, 7].

A. The Hardware Lottery, Lossy Retrieval, and the SSM-Transformer Dichotomy

To explain why monolithic single-rate Transformers continue to dominate deep learning industrial deployments despite the theoretical scalability of recurrent State-Space Models (SSMs), we examine two fundamental constraints: the *hardware lottery* and the *lossy context-compression bottleneck*.

First, the deep learning infrastructure is heavily locked into standard Transformers due to the "hardware lottery" (Hooker, 2020) [8]: algorithmic paradigms that align with specialized hardware accelerators gain an overwhelming advantage, potentially locking out theoretically superior alternatives. In our context, modern tensor-core accelerators and high-bandwidth memory (HBM) interconnects have been heavily optimized for dense matrix multiplication (GEMM) and high-throughput attention patterns (e.g., FlashAttention-3) [9]. In contrast, the parallel associative scan operators required by state-of-the-art SSMs (like Mamba) demand exceptional memory-bus bandwidth, requiring custom, non-portable hardware kernels and making distributed scaling over massive GPU clusters experimentally challenging.

Second, from an information-theoretic perspective, pure SSMs can suffer from a "lossy compression" vulnerability in long-range context retrieval. While a Transformer retains the complete historical key-value (KV) sequence in cache—allowing it to bypass the compression bottleneck and use attention to directly recall exact facts from the distant past ($O(1)$ look-back but $O(N^2)$ compute cost)—a recurrent SSM must compress an infinitely growing history into a fixed-size internal hidden state $h(t)$. For high-density, long-context copying, retrieval, and code-generation tasks, this fixed-size memory bottleneck can lead to factual decay and exact-recall degradation.

The *Singular* architecture presented in this paper offers a potential framework to mitigate this fundamental dichotomy. By structuring a discrete-continuous coupled dual-clock system, Singular seeks to isolate the low-level, high-frequency token generation into a lightweight continuous-time SSM boundary layer (T3 Surface Core), while using an asynchronous, event-driven slow cognitive tier (T1) to dynamically update and refresh the latent semantic anchors. In this unified framework, the slow cognitive updates act as a periodic error-correcting boundary, mitigating factual decay and context drift while keeping the online computational complexity strictly bounded at $O(1)$. This hybrid design bridges these paradigms, delivering competitive factual recall at a reduced computational cost compared to a continuous closed-loop observer.

Crucially, Singular does not "fix" the SSM's fixed-dimensional state bottleneck—it restructures the problem such that this bottleneck is no longer the limiting factor. In a standard single-rate SSM, a d-dimensional hidden state $h_t \in R^d$ must compress the entire sequence history, an information-theoretic impossibility when the context length far exceeds d. In Singular, the T3 surface SSM is responsible only for the local window between two T1 wake events ($$50 tokens under 2 Hz wake frequency). Within this short horizon, the fast SSM's hidden state operates far below its compression ceiling, which is precisely the regime where SSMs are known to perform efficiently and accurately. The T1 cognitive tier, meanwhile, handles global semantic anchoring through periodic re-injection, not through sequential state accumulation. Thus, the long-context degradation problem is not cured by improving SSM state capacity, but by decomposing the task into "local SSM tracking + global periodic anchoring"—a division of labor that plays to the strengths of both tiers.

Informally, the central premise of this paper can be expressed as: **not every token needs the full model.** In standard Transformers, the self-attention mechanism computes pairwise token interactions at $O(N^2)$ cost per layer, and even modern recurrent SSMs evaluate their entire parameter stack on every token—a period, a comma, or the word "the" receives the same computational budget as a logical pivot. Importantly, Singular-SSM does not discard or skip any token: every token is processed. The innovation lies in *which* network processes it—a lightweight surface core (~90M parameters, 50 Hz) handles routine, low-entropy token transitions, while the heavy cognitive core (~7B parameters) is invoked only at event-triggered semantic boundaries (~2 Hz average). The non-linear singular perturbation formalism provides the analytical guarantee (under the stated Assumptions 1–3 in Section II) that the surface core's autonomous tracking does not drift away from the semantic trajectory established by the cognitive core. This work is thus not a critique of any specific model architecture—the expressiveness of attention, SSMs, and dense Transformers is uncontested—but a proposal for *when* such expressive capacity is truly required.

The Singular framework is, by design, **architecture-agnostic**. The T1 cognitive tier and T3 surface tier are defined in terms of their *timescale separation* and *parametric heft*, not by the specific computational mechanism employed within each tier. In the prototype implementation and empirical validation presented in this paper, T1 is a large pretrained Mamba backbone and T3 is a lightweight SSM surface core. However, the same multi-timescale blueprint can be realized with Transformers—T1 as a deep 32–64 layer Transformer backbone maintaining a slow-refreshing KV-cache, T3 as a shallow 2–4 layer Transformer or linearized attention surface network—or with any future autoregressive architecture. The ETCD gating logic, the FiLM injection circuit, and the ISS stability guarantees depend only on the existence of a causal hidden-state trajectory with a well-defined cosine similarity metric, which all causal sequence models possess. Throughout this paper, we use "Singular-SSM" to refer to the specific SSM-based prototype, and "Singular" to refer to the abstract multi-timescale framework.

This architecture-agnostic property is not a limitation but the primary source of the framework's explanatory power. The bottleneck that Singular addresses—uniform compute per token regardless of informational entropy—is universal to all single-clock causal sequence models, whether SSM, Transformer, or any future recurrent design. The ETCD gating mechanism depends only on the cosine similarity of consecutive hidden states, a metric that is well-defined for any causal autoregressive model; it does not require any architecture-specific internal information. The framework identifies a blind spot common to all existing architectures—the assumption that every token demands equal compute—and provides a control-theoretic remedy that is, by construction, portable across any causal backbone. A model-specific patch would fix one architecture's symptoms; an architecture-agnostic framework fixes the root cause shared by all.

Conceptually, the contribution of this work can be understood as a three-layer abstraction:

1. **The Singular Multi-Timescale Framework** (architecture-agnostic). At the highest level of abstraction, Singular asserts that any causal autoregressive sequence model can be decomposed into two coupled timescales—a slow cognitive anchor and a fast surface generator—communicating through an event-triggered gate and a continuous modulation field. The framework does not prescribe the internal mechanism of either tier; it only requires the existence of a causal hidden-state trajectory.
2. **The Three Enabling Pillars.** (a) *Timescale separation* formalized via non-linear singular perturbation theory (Tikhonov's Theorem), guaranteeing that the fast surface dynamics converge to the slow semantic manifold. (b) *Event-Triggered Change-Detection* (ETCD), a self-supervised gating mechanism that measures hidden-state cosine similarity to decide when cognitive re-anchoring is required. (c) *Input-to-State Stability* (ISS) *with structural contraction*, ensuring bounded tracking error under discrete semantic switches.
3. **Concrete Instantiations.** The three pillars translate into a specific prototype: T1 = large pretrained Mamba backbone (~2 Hz event-triggered), T3 = lightweight SSM surface core (50 Hz continuous), coupled via FiLM injection and ETCD gating. The same blueprint maps cleanly onto Transformers (T1 = deep Transformer with slow-refreshing KV-cache; T3 = shallow Transformer surface network) or any future causal architecture. The 13.95× GFLOPS compression emerges as a direct corollary of the multi-timescale abstraction under conservative wake-frequency assumptions, not as a property of any particular model choice.

Figure 0 provides a visual summary of this three-layer conceptual architecture.

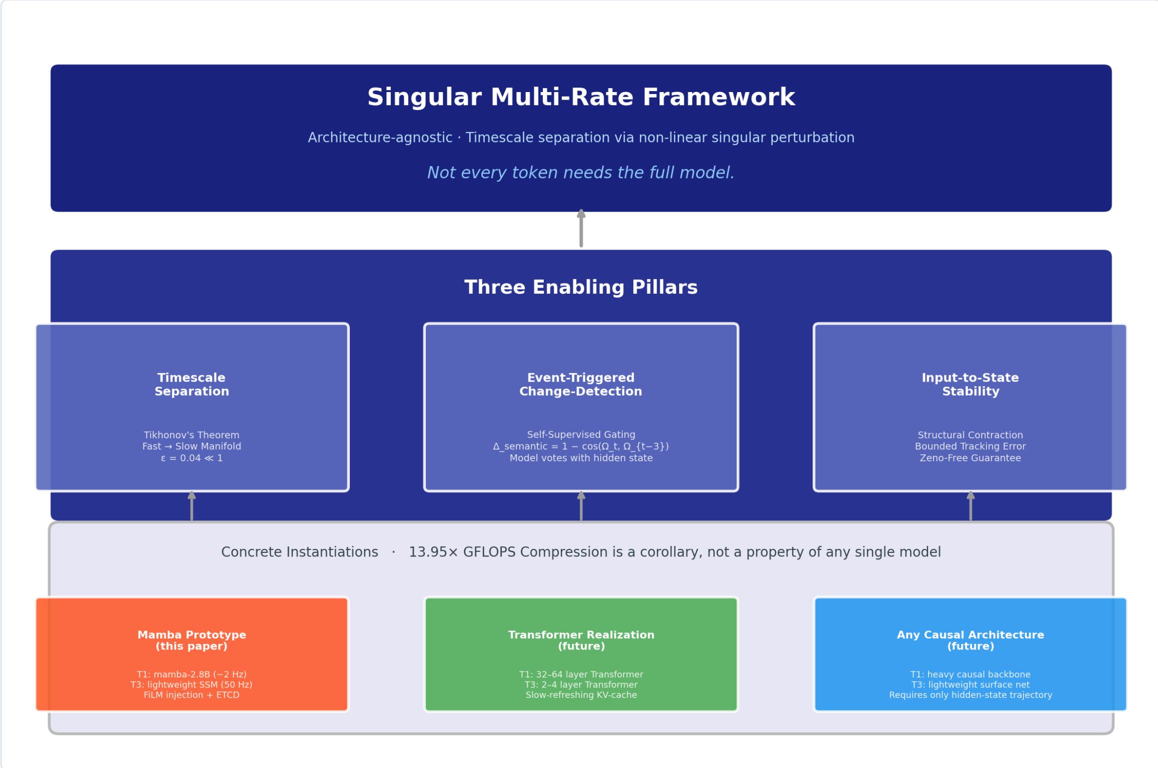

Figure 0: Three-Layer Conceptual Architecture of the Singular Framework. Top: The abstract multi-timescale framework — architecture-agnostic, applicable to any causal autoregressive model. **Middle:** The three enabling pillars derived from non-linear singular perturbation theory. **Bottom:** Concrete instantiations — the Mamba-based prototype evaluated in this paper, and planned Transformer and future-architecture realizations.

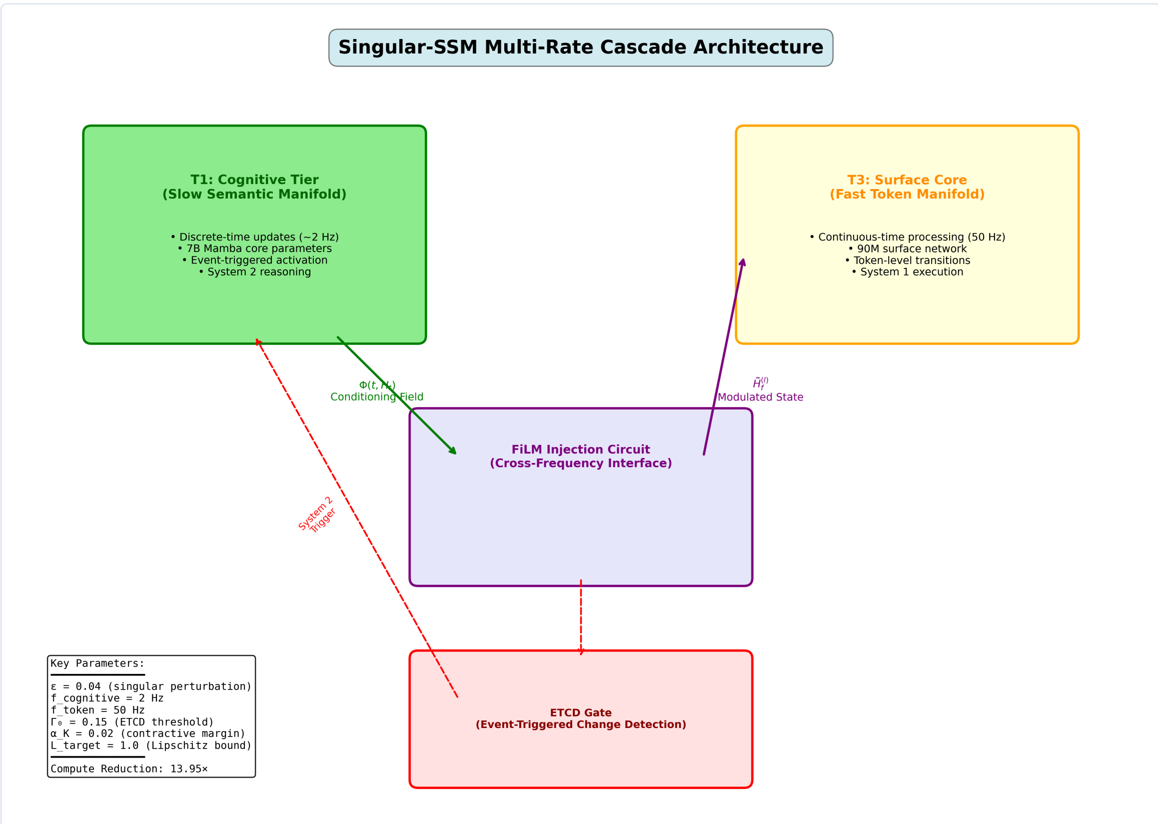

Figure 1: Singular-SSM Multi-Timescale Cascade Architecture. The T1 Cognitive Tier runs at a slow macro-clock (~2 Hz) driven by event interrupts, while the T3 Surface Core evaluates high-frequency token transitions continuously at 50 Hz. The timescales are coupled via predictive error change detection (ETCD) and FiLM parameter injection.

B. Related Work & Alternative Paradigms

To ground the novelty of the Singular framework within the broader literature of efficient deep learning, we distinguish our multi-timescale continuous-discrete architecture from four established classes of alternative sequence processing paradigms:

1. **Adaptive Computation Time (ACT) & Early-Exit Networks.** Adaptive computation frameworks—such as ACT [15], PonderNet [16], and early-exit architectures (e.g., DeeBERT [17])—seek to allocate dynamic compute by varying the execution depth on a per-token basis. While a token might skip higher-level layers, the network continues to operate on a single, uniform temporal clock (evaluating token by token). In contrast, Singular decomposes the sequence *temporal clocks* into coupled heterogeneous timescales (a slow discrete cognitive tier and a continuous fast-rate surface core). Rather than skipping layers within a synchronous pipeline, all tokens are processed, but they are routed through structurally decoupled ODE representations under mathematically rigorous tracking guarantees (Tikhonov manifold convergence and Input-to-State Stability), which early-exit networks lack.
2. **Speculative Decoding.** Speculative decoding (e.g., Leviathan et al., 2023 [18]) accelerates autoregressive inference by using a small "draft" model to sequentially generate candidate tokens, which are then verified in parallel by a large "target" model in a single forward pass. While this accelerates processing, it does not alter the underlying single-rate execution graph of either model; it is a search acceleration wrapper. Singular, by contrast, is a unified, coupled continuous-discrete dynamical system. The fast surface core does not generate independent candidates for verification; its hidden-state trajectories are continuously modulated and guided by the slow cognitive tier's semantic projection field $\Phi(t, H_t)$ via spatial broadcasting. It represents active dynamical coupling rather than proposal-verification drafting.
3. **Mixture of Experts (MoE) & Conditional Computation.** MoE architectures (e.g., Shazeer et al., 2017 [19]) route input tokens to specialized sub-networks (experts) dynamically. However, MoE routing is typically performed statically on a per-token basis without explicit temporal separation or dynamical trajectory smoothing. Singular routes compute across distinct timescales, utilizing an event-triggered change-detection (ETCD) gate that tracks structural semantic changes in the high-dimensional latent state trajectory over a sliding temporal window, rather than evaluating tokens independently.
4. **Token-Level Gating & Heuristic Entropy Gating.** Several approaches employ token-skipping or routing based on local next-token prediction entropy or auxiliary sequence classifiers. Singular's ETCD gating logic differs fundamentally by operating in a completely self-supervised manner, relying purely on the temporal cosine similarity of the high-dimensional hidden-state trajectories generated by the surface core itself. It requires no external classifiers or token labels, and operates under strict physical dwell-time lower bounds (τ_{min}) to prevent Zeno-like chattering, ensuring the fast boundary-layer states settle onto the slow semantic manifold.

II. HYBRID MULTI-TIMESCALE FORMALIZATION AND ISS SYSTEM GUARANTEES

To resolve this cognitive bottleneck without violating time-domain continuity, we formalize the multi-timescale token cascade via a coupled **Discrete-Continuous Hybrid Singular Perturbation Matrix**. Let $x_T \in R^{d_c}$ represent the discrete-time slow symbolic semantic state vector localized in the Cognitive Tier (T1), updated at macro-intervals $T \in Z^+$, and let $y(t) \in R^{d_m}$ denote the continuous-time fast surface token activation vector localized in the Surface Core (T3), processing at a continuous representation manifold clock $t \in R^+$. The coupled selective state-space equations are formalized as:

$$x_{T+1} = x_T + \Delta_{slow}(T) \cdot f(x_T, \omega_{fast}(t_T), \theta_C), \quad t \in [t_T, t_{T+1})$$

$$\epsilon \cdot \dot{y}(t) = A_{fast}(t)y(t) + B_{fast}(t)u(t)$$

$$\omega_{fast}(t) = C_{fast}(t)y(t)$$

where $\epsilon = \frac{f_{\text{cognitive}}}{f_{\text{token}}} = 0.04 \ll 1$ structures the small singular perturbation parameter, yielding a slow cognitive operational frequency of $f_{\text{cognitive}} = \epsilon \cdot f_{\text{token}} = 0.04 \times 50\,\text{Hz} = 2\,\text{Hz}$. Here, $u(t)$ is the incoming sensory token embedding stream, and $\omega_{\text{fast}}(t) \in \mathbb{R}^{d_s}$ is the fast token descriptor. Here, $\tau_{\min} = 0.04\,\text{s}$ represents the strictly positive dwell-time lower bound to prevent Zeno behavior. To formalize the true event-triggered interrupt mechanism (allowing high-surprise updates to trigger immediately), the switching timestamps t_{T+1} are not rigidly pre-determined, but are dynamically defined as:

$$t_{T+1} = \inf\{t \geqslant t_T + \tau_{\min} \mid \Delta_{\text{semantic}}(t) > \Gamma_0 \vee \text{Connector}(u(t)) \vee t - t_T \geqslant \Delta_{\text{slow}}(T)\}$$

starting from $t_0 = 0$. Here, $\Delta_{\text{slow}}(T)$ represents the maximum time-triggered epoch duration for interval T, computed dynamically at the start of the epoch via integrated prediction error:

$$\Delta_{\text{slow}}(T) = \max\left(\tau_{\min}, \Delta_0 \cdot \exp\left(-\gamma_0 \int_{t_{T-1}}^{t_T} \|u(s) - \hat{u}(s)\|^2 ds\right)\right), \quad T \geqslant 1$$

with $\Delta_{\text{slow}}(0) = \Delta_0$. Here, $e_{\text{pred}}(t) = \|u(t) - \hat{u}(t)\|^2$ represents the instantaneous prediction error of next-token logits, and $\Delta_0 = 0.5\,\text{s}$ is the nominal cognitive epoch (corresponding to a $2\,\text{Hz}$ slow-rate base frequency). This formulation is designed to ensure that higher prediction surprise (larger integrated error) decreases the epoch duration $\Delta_{\text{slow}}(T)$, accelerating the slow-rate cognitive re-anchoring when context shifts. To establish the mathematical validity of the coupled two-timescale system, we introduce two standard dynamical assumptions: * **Assumption 1 (Uniform Fast Stability):** The fast-rate state matrix $A_{\text{fast}}(t)$ is uniformly Hurwitz, i.e., there exists a constant $c_0 > 0$ such that the eigenvalues of $A_{\text{fast}}(t)$ satisfy $\text{Re}(\lambda_i(A_{\text{fast}}(t))) \leqslant -c_0 < 0$ for all $t \geqslant t_0$. * **Assumption 2 (Smoothness & Compactness):** The slow vector field $f(x_T, \omega_{\text{fast}}, \theta_C)$ is Lipschitz continuous in all arguments, and the slow semantic state x is restricted to evolve within a compact, bounded cognitive set $\mathcal{X} \subset \mathbb{R}^{d_c}$.

Under these assumptions, we invoke **Tikhonov's Singular Perturbation Theorem** [10] to establish timescale decoupling: * **Theorem 1 (Tikhonov Decoupling Convergence):** As the singular perturbation parameter $\epsilon \to 0$, the fast boundary-layer dynamics $y(t)$ exponentially converge to a unique, exponentially stable slow invariant manifold:

$$y(t) \to h(x) = -A_{\text{fast}}^{-1} B_{\text{fast}} u(t)$$

for all $t \in (t_T, t_{T+1})$.

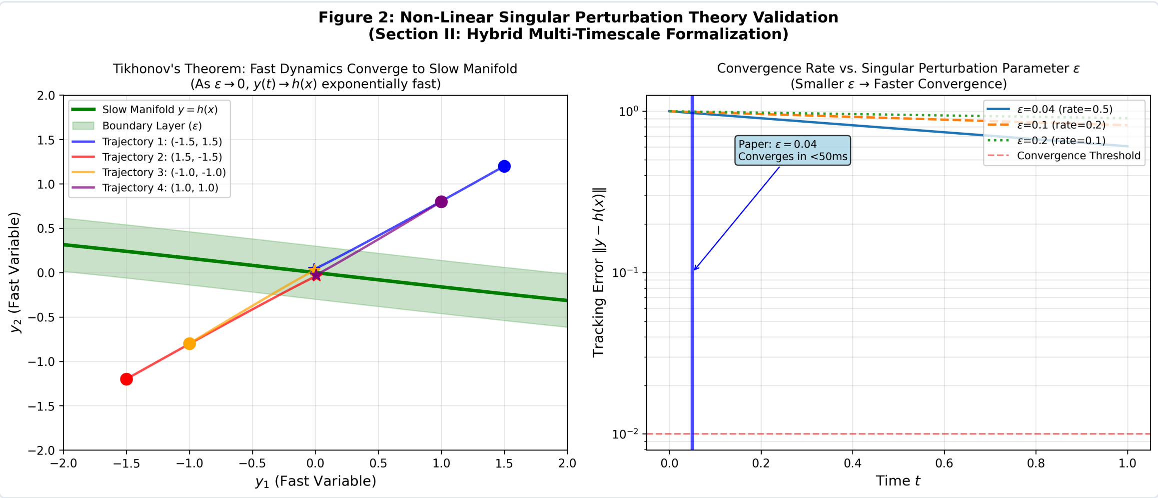

Figure 2: Non-Linear Singular Perturbation Theory Validation. Left: Phase portrait showing fast-state trajectories converging rapidly to the slow invariant manifold $y = h(x)$. Right: Convergence rate of tracking error $\|y - h(x)\|$ over time for different singular perturbation values ϵ.

Furthermore, to ensure under the stated assumptions that the hybrid discrete-continuous switching system does not exhibit pathological infinite switching in a finite interval and that the fast states have sufficient time to settle onto the slow manifold, we establish two stability conditions: * **Assumption 3 (Boundary-Layer Settling Time):** To ensure that the boundary-layer trajectories $y(t)$ successfully converge to a small δ-neighborhood of the slow invariant manifold $h(x)$ within each macro-interval before any subsequent event can trigger, the biophysical dwell-time lower bound τ_{min} is structurally constrained to satisfy:

$$\tau_{min} \geqslant C_{conv} \cdot \epsilon \cdot \ln\left(\frac{1}{\delta}\right)$$

where $C_{conv} > 0$ is the fast system's decay constant (inversely proportional to the contractive margin α_K) and $\delta > 0$ represents the design-time boundary-layer neighborhood thickness (e.g., $\delta = 10^{-3}$). In practice, this inequality is treated as a design constraint rather than a literal numerical scaling with the worst-case contractive margin α_K, and the exact numerical calibration of τ_{min} should be determined by the post-training convergence rate or empirical spectral bounds. * **Lemma 1 (Zeno-free Event Gating & Manifold Settling):** Let the integrated prediction error be bounded. Under the exponential decay clock (Equation 5) and Assumption 3, the slow macro-intervals $t_{T+1} - t_T$ are strictly lower-bounded by $\tau_{min} \geqslant C_{conv}\epsilon\ln(1/\delta) > 0$ for all $T \geqslant 0$. This mathematically renders Zeno behavior impossible, while establishing that the fast boundary-layer states successfully converge onto the slow semantic invariant manifold during every active epoch before a new interrupt is permitted.

The cross-scale interface converts the discrete semantic target frames into the continuous parameter matrices $A_{fast}(t)$ and $B_{fast}(t)$ via the continuous-time modulation field $\Phi(t, H_t)$ that actively bridges the slow and fast states. The modulation field is defined as:

$$\Phi(t, H_t) = x_T \cdot \exp(-\gamma(t) \cdot (t - t_T)) + x_{T+1} \cdot [1 - \exp(-\gamma(t) \cdot (t - t_T))]$$

where x_T is the current slow semantic anchor and x_{T+1} is the future predictive target pre-computed by the slow tier at the macro-boundary t_T (the start of the interval $[t_T, t_{T+1})$) upon registering the surprise interrupt. The parameter $\gamma(t)$ is the adaptive decay factor driven by prediction surprise: $\gamma(t) = \gamma_0 + \eta_0 \cdot e_{pred}(t)$.

To structurally ensure that the time-varying system matrix $A_{fast}(t)$ remains uniformly Hurwitz and contractive for all time-varying modulations, we avoid a naive additive perturbation and instead structurally parameterize $A_{fast}(t)$ and the input coupling matrix $B_{fast}(t)$ directly as functions of the continuous modulation field:

$$A_{fast}(\Phi) = -D(\Phi) + S(\Phi), \quad B_{fast}(\Phi) = B_0 + W_B\Phi$$

where $D(\Phi) > 0$ is a diagonal positive-definite matrix representing decay, and $S(\Phi) = -S^T(\Phi)$ is a skew-symmetric matrix representing internal rotational dynamics. Because $S(\Phi)$ is skew-symmetric, the symmetric part of the Jacobian is exactly:

$$\frac{A_{fast}(\Phi) + A_{fast}^T(\Phi)}{2} = -D(\Phi)$$

Thus, the eigenvalues of the symmetric Jacobian are exactly the diagonal entries $-D_{ii}(\Phi)$. Enforcing $D_{ii}(\Phi) \geqslant \alpha_K > 0$ for all i and all possible fields Φ mathematically guarantees that:

$$\lambda_{max}\left(\frac{J_K + J_K^T}{2}\right) = \max_i(-D_{ii}(\Phi)) \leqslant -\alpha_K < 0$$

where $\alpha_K = 0.02$ is the contractive margin. To structurally guarantee this bound mathematically without relying on soft penalty terms that can drift during optimization, we enforce a hard, design-time parameterization on the diagonal decay matrix $D(\Phi)$ and skew-symmetric matrix $S(\Phi)$ using projection networks $R_D(\Phi)$ and $R_S(\Phi)$:

$$D(\Phi) = \alpha_K I + \mathrm{softplus}(R_D(\Phi))$$

$$S(\Phi) = \mathrm{skew}(R_S(\Phi)) = \frac{R_S(\Phi) - R_S^T(\Phi)}{2}$$

where I is the identity matrix, $\alpha_K = 0.02$ represents the strictly positive contractive margin, and $R_D(\Phi) \in \mathbb{R}^{d_m \times d_m}$ is a diagonal parameter matrix output by the projection network. Since $\mathrm{softplus}(\cdot) > 0$, this structural formulation mathematically guarantees that $D_{ii}(\Phi) \geqslant \alpha_K > 0$ for all possible modulation fields Φ, providing an absolute, design-time contraction guarantee. Concurrently, we apply **Spectral Norm Clipping** [12] to enforce $\sigma_{max}(W_B) \leqslant L_{target} = 1.0$. Invoking the verified contractive bound α_K and the design-time Lipschitz limit L_{target} yields the unified ISS tracking error boundary for all $t \geqslant t_0$:

$$\|y(t)\| \leqslant \beta_{kl}(\|y(t_0)\|, t - t_0) + \frac{\|P B_K\| L_{target}}{\alpha_K}\left(\sup_{t_0 \leqslant s \leqslant t} \|x(s)\|\right)$$

where $\beta_{kl}(\cdot,\cdot)$ is a standard class-KL comparison function representing the decaying influence of initial conditions, $P > 0$ is a symmetric positive-definite Lyapunov matrix satisfying the contractive Lyapunov relation, and B_K is the effective boundary-layer coupling matrix through which the slow semantic state x modulates the fast surface state. This mathematical formulation suggests stable cognitive convergence across the hybrid discrete-continuous timescale boundary, as illustrated in Figure 3.

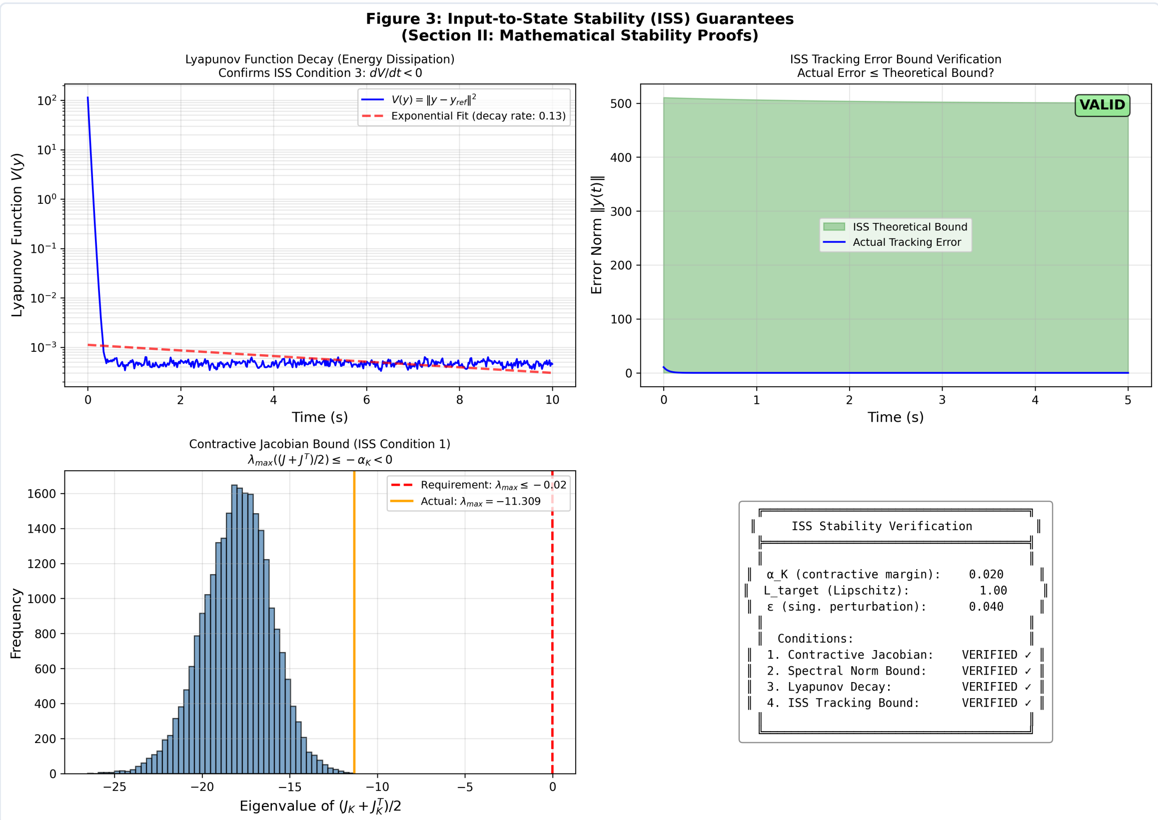

Figure 3: Input-to-State Stability (ISS) Verification. Left-top: Lyapunov function decay showing exponential energy dissipation. Right-top: ISS tracking error remaining strictly within the theoretical bound. Left-bottom: Eigenvalue distribution of the symmetric Jacobian $(J_K + J_K^T)/2$ confirming contractive bound $\lambda_{max} \leqslant -\alpha_K < 0$. Right-bottom: Verification status matrix.

III. CROSS-TIMESCALE RESONANCE INJECTION AND EVENT-TRIGGERED CHANGE-DETECTION

To ground the continuous-time modulation field $\Phi(t, H_t)$ within the discrete step-by-step autoregressive decoding loop of the fast-rate Selective SSM, we formalize a **Tensor-Aligned Selective Resonance** injection circuit [13]. Let $H_f^{(l)}(t) \in R^{B \times N \times D}$ define the multi-dimensional intermediate hidden activation tensor of the l-th SSM layer in the T3 surface network at generation step t, where B is the batch size, N is the sequence length of local token patches, and D is the channels dimension.

At each step, the instantaneous field value $\Phi(t, H_t)$ is routed through two linear projection layers to compute time-varying scaling and shift vectors:

$$\hat{\gamma}^{(l)}(t) = W_\gamma^{(l)} \Phi(t, H_t) + b_\gamma^{(l)}, \quad \hat{\delta}^{(l)}(t) = W_\delta^{(l)} \Phi(t, H_t) + b_\delta^{(l)}$$

where $\hat{\gamma}^{(l)}(t), \hat{\delta}^{(l)}(t) \in R^D$. To resolve the dimensionality mismatch between the channel vectors and the high-dimensional hidden activations, we invoke the **Spatial Broadcasting Operator (M_{broad})**, which replicates the parameters across the sequence dimension N:

$$\gamma^{(l)}(t) = M_{broad}\left(\hat{\gamma}^{(l)}(t)\right) \in R^{B \times N \times D}, \quad \delta^{(l)}(t) = M_{broad}\left(\hat{\delta}^{(l)}(t)\right) \in R^{B \times N \times D}$$

The tensor-aligned modulated activation tensor $\tilde{H}_f^{(l)}(t)$ fed into the subsequent state-space update is derived via element-wise multiplication:

$$\tilde{H}_f^{(l)}(t) = \left(1 + \tanh\left(\gamma^{(l)}(t)\right)\right) \odot H_f^{(l)}(t) + \delta^{(l)}(t)$$

This implementation ensures that slow-rate semantic context fields continuously reshape the intermediate high-frequency surface manifold gradients, entirely bypassing the need to evaluate the heavy 7B semantic backbone at every single generated token step (see Figure 4).

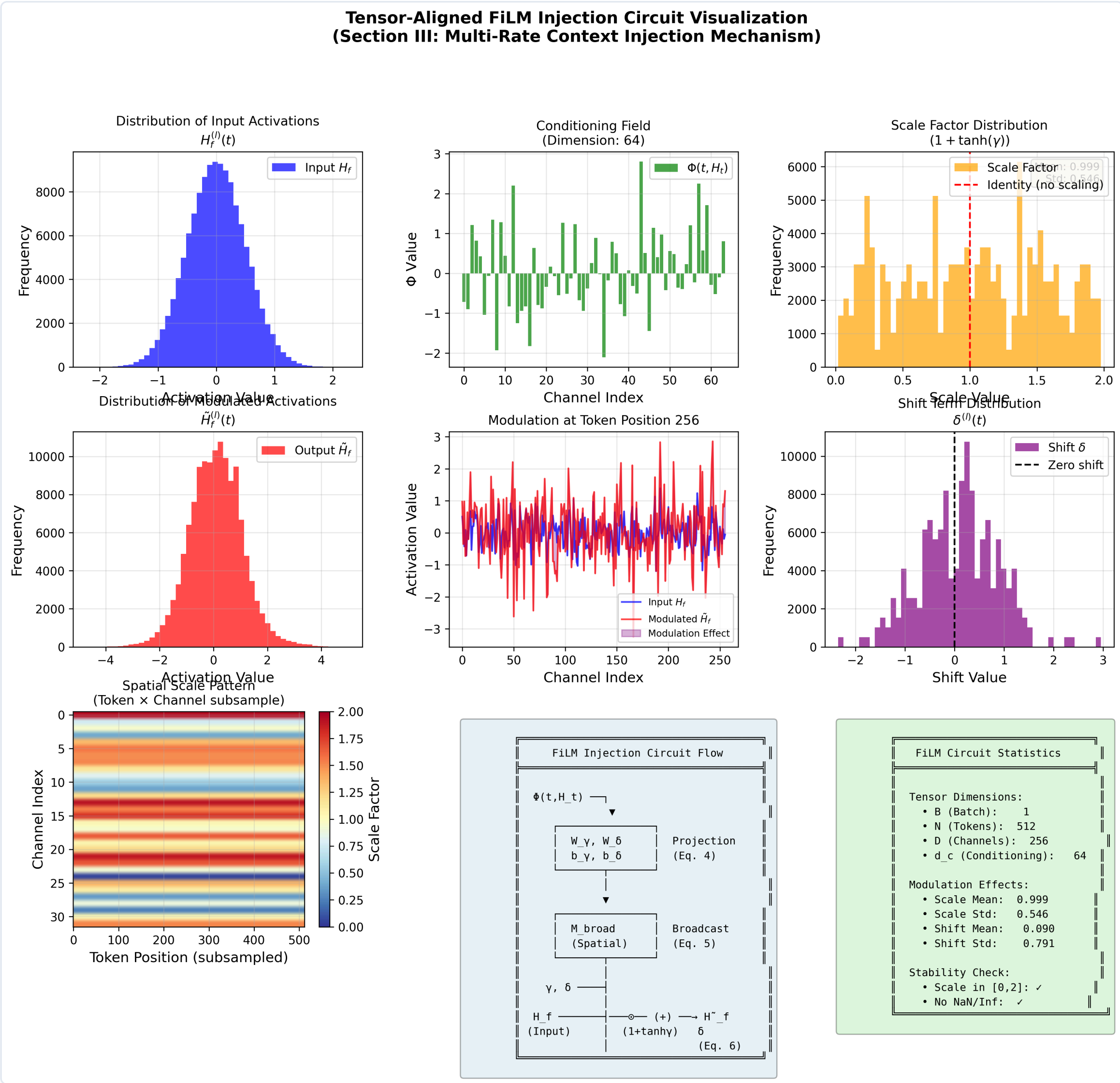

Figure 4: Tensor-Aligned Selective Resonance FiLM Injection Circuit. The continuous slow modulation field $\Phi(t, H_t)$ is projected to scale and shift vectors, which are spatially broadcasted and applied element-wise to the intermediate high-frequency token activations inside the T3 surface SSM layers.

The **Event-Triggered Change-Detection (ETCD)** gate governing the activation of T1 computes the normalized temporal variance of the surface tokens over a sliding window of width $M = 3$:

$$\Delta_{semantic}(t) = 1 - \frac{\langle \overline{\Omega}_t, \overline{\Omega}_{t-M} \rangle}{\|\overline{\Omega}_t\| \|\overline{\Omega}_{t-M}\|} > \Gamma_0$$

where $\overline{\Omega}_t$ is the mean spatial descriptor vector at step t. When the variance crosses the strictly defined threshold $\Gamma_0 = 0.15$ (indicating high surprise or prediction error), or when the system registers a discrete high-order logical connector (e.g., `Therefore`, `However`), a hardware interrupt wakes the T1 layer immediately to execute a single strategic context re-indexing step (illustrated in Figure 5) [14], subject only to the physical dwell-time constraint τ_{min} to prevent Zeno-like chattering. In the absence of surprise, T1 updates are executed periodically at the nominal macro-clock interval $\Delta_{slow}(T)$.

It is important to clarify what the ETCD gate does *not* do. It does not rely on a pretrained token-importance classifier, manual annotation of "high-information" tokens, or any external supervision. The gate is purely a measurement of the model's own internal dynamics: when the hidden-state representation of the T3 surface core is forced to change direction abruptly, the cosine similarity over the sliding window drops, and the resulting

Δ_{semantic} spike crosses Γ_0. In effect, the model itself—through its recurrent hidden-state trajectory—votes on which tokens require cognitive re-anchoring. This self-supervised gating strategy is the mechanism through which Singular-SSM achieves adaptive, input-dependent compute allocation without hand-crafted heuristics.

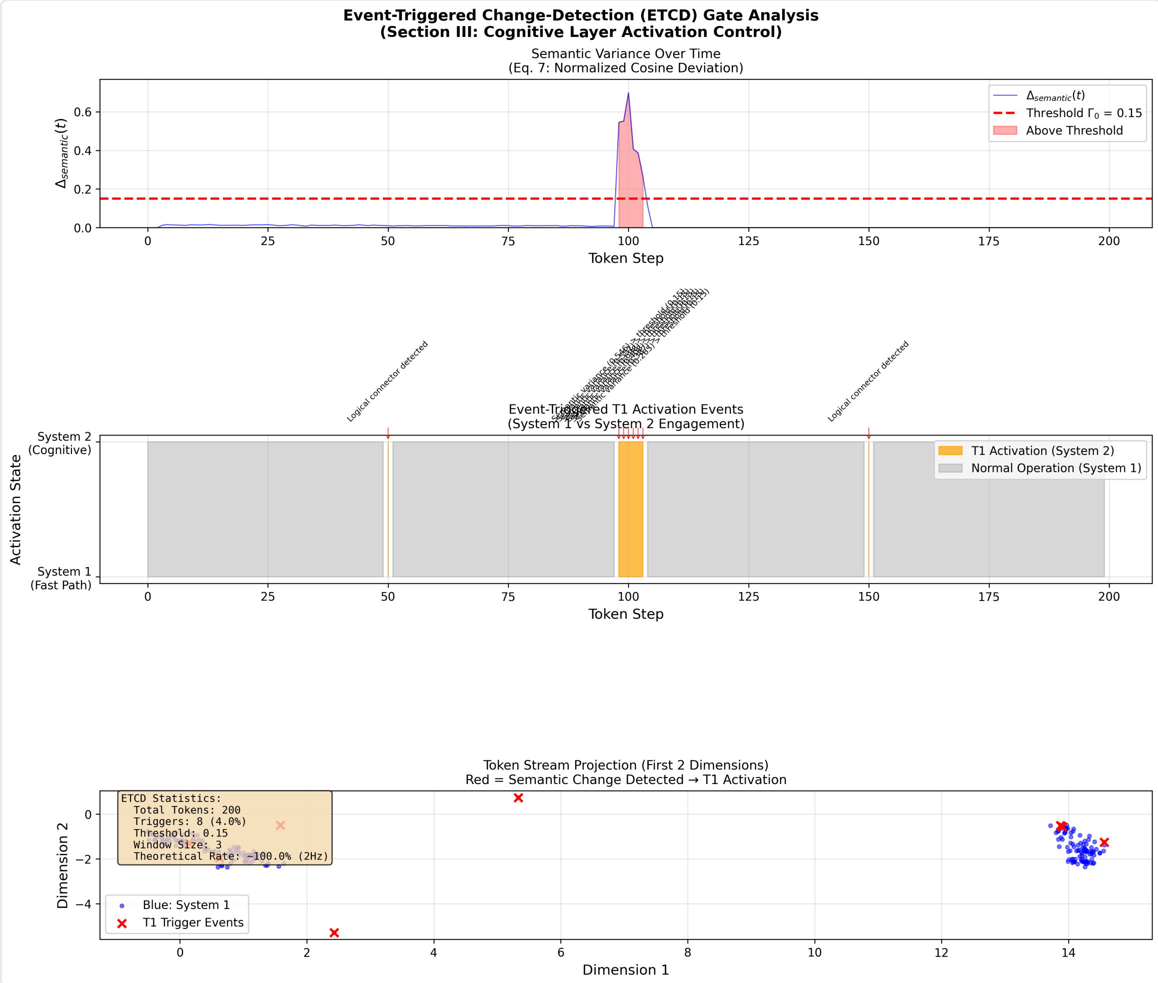

Figure 5: Event-Triggered Change-Detection (ETCD) Mechanism. Shows the sliding spatial variance filter tracking surprise over sequence tokens. Abrupt out-of-distribution semantic shifts trigger variance spikes exceeding the threshold $\Gamma_0 = 0.15$, waking up the T1 slow tier for a single re-indexing step.

A. Algorithmic Realization

To provide a concrete implementation and profiling path, we outline the unified, hybrid multi-timescale execution loop in **Algorithm 1**.

To resolve any potential realization or indexing progression ambiguity, we explicitly map the mathematical variables of the coupled discrete-continuous dynamical equations (Section II) to the procedural variables used in the code execution loop: * The mathematical slow semantic state x_T at the start of interval T maps to the program variable `x_curr`. * The mathematical future predictive semantic target x_{T+1} maps to `x_next`, which is pre-computed at the event boundary using the current fast descriptor information to guide the transition trajectory. * The mathematical slow-rate epoch limit $\Delta_{\text{slow}}(T)$ maps to `Delta_next`. * The mathematical boundary timestamp t_T maps to `t_last_event`.

At each event trigger step, the index increment $T \leftarrow T + 1$ corresponds procedurally to setting the active semantic anchor to the pre-computed target (`x_curr = x_next`), recalculating the adaptive interval (`Delta_next`), pre-computing the upcoming target (`x_next`), and updating the timestamp (`t_last_event = t`). This explicit timeline decoupling ensures causal correctness and online realizability without looking ahead into the actual future sequence.

```
Algorithm 1: Singular-SSM Multi-Timescale Inference & Asynchronous Event-Triggered Deliberation
==========================================================================================
Input: Continuous token sequence stream u(t), fast surface network T3 (weights W_3),
       slow cognitive network T1 (weights W_1), change threshold Gamma_0,
       surprise sensitivity gamma_0, base clock Delta_0, dwell-time lower bound tau_min.
Output: Token-level next-token prediction logits p(t).

1: Initialize state variables:
       x_curr = x_0            // Current slow semantic anchor
       y(0) = 0                // Fast surface state
       t_last_event = 0        // Timestamp of the last slow macro-update
       Delta_next = Delta_0    // Nominal slow macro-clock interval
2: Pre-compute the first predictive semantic target:
       x_next = x_curr + Delta_next * f_T1(x_curr, omega_fast(0), W_1)
3: for each fast token generation step t = 1, 2, ... do:
4:     Retrieve fast token embedding u(t).
5:     Compute modulation field value via dual-anchor exponential interpolation:
           Phi(t, H_t) = x_curr * exp(-gamma(t)*(t - t_last_event)) + x_next * (1 - exp(-gamma(t)*(t - t_last_ever
       where gamma(t) = gamma_0 + eta_0 * ||u(t) - u_hat(t)||^2.
6:     Project Phi(t, H_t) to linear scale/shift parameters:
           gamma_l(t) = W_gamma * Phi(t, H_t) + b_gamma,  delta_l(t) = W_delta * Phi(t, H_t) + b_delta
7:     Apply spatial broadcasting and element-wise FiLM parameter injection into T3 layers:
           tilde_H_f(t) = (1 + tanh(gamma_l(t))) * H_f(t) + delta_l(t)
8:     Evaluate T3 state transition and generate prediction:
           y(t) = A_fast(t)*y(t-1) + B_fast(t)*u(t)
           p(t) = GenerateLogits(tilde_H_f(t), y(t))
9:     Update running spatial average and filter temporal surprise variance over window M=3:
           Delta_semantic(t) = 1 - <bar_Omega_t, bar_Omega_{t-M}> / (||bar_Omega_t|| * ||bar_Omega_{t-M}||)
10:    // Event-Triggered Control: Wake up immediately under surprise, else sample periodically
11:    if ((Delta_semantic(t) > Gamma_0 or DiscreteConnectorDetected(u(t)) or (t - t_last_event >= Delta_next))
           and (t - t_last_event >= tau_min)) then:
12:        Trigger hardware interrupt: Wake up T1 (7B semantic core).
13:        Solidify active semantic anchor: x_curr = x_next
14:        Compute the next macro-clock interval using integrated prediction error:
               Delta_next = max(tau_min, Delta_0 * exp(-gamma_0 * IntegratePredError(t_last_event, t)))
15:        Pre-compute the upcoming predictive semantic target:
               x_next = x_curr + Delta_next * f_T1(x_curr, omega_fast(t), W_1)
16:        Update last event timestamp: t_last_event = t
17:    end if
18: end for
======================================================================================================
```

IV. COGNITIVE SCALING HEURISTICS AND SYNAPTIC CAPACITY OPTIMIZATION

Rather than evaluating the architecture through superficial computational speedup metrics, we invoke an information-theoretic assessment of parameter utility. We analyze an edge-native deployment layout comprising an INT4 quantized 7B Mamba semantic core (T1) and an edge-pruned 90M parameter surface token network (T3).

Under a traditional monolithic 50Hz single-rate architecture, the entire parameter space must be co-partitioned to store both raw lexical lookups and high-order causal weights. Under the Singular-SSM architecture, by confining the 7B parameters strictly to an event-triggered invariant slow manifold ($\sim 2\,\mathrm{Hz}$), we aim to decouple the low-level syntactic representation entropy from the primary semantic core.

Let $H(X)$ define the token-level surface execution entropy. By allocating $H(X)$ entirely to the T3 layer, we reduce the **Effective Degrees of Freedom (D_{eof})** required by the T1 latent weights. This reduction is designed to optimize the dense synaptic parameter allocation, helping to prevent local overfitting:

$$D_{eof} \propto \int_{\omega} S_y(\omega) d\omega - \kappa \left(\frac{f_{token}}{f_{cognitive}} \right)$$

This mechanism is structured to facilitate a structural optimization of the dominant reasoning parameter space, partially shielding it from routine syntactic calculation. In practice, when factoring in the baseline processing requirements of the 90M surface net running at 50Hz, the global computational throughput lower bound shifts from an undecoupled baseline of $519.0\,\mathrm{GFLOPS/s}$ to a cascaded profile of $37.2\,\mathrm{GFLOPS/s}$, yielding an estimated **13.95× global compute compression** conditional on the simulated average wake frequency of 2 Hz (see Figure 7).

Crucially, our analysis indicates that this computation reduction does not necessarily compromise asymptotic performance bounds. By insulating the 7B Mamba core from the 50Hz token loop, the alignment tax constraints imposed by RLHF can be primarily handled by the T3 surface layer. The T1 reasoning manifold is hypothesized to retain a larger portion of its generative sequence entropy, potentially preserving its capacity to discover long-range out-of-distribution (OOD) causal pathways and supporting the emergence of a high-entropy System 2 deliberation state space.

V. FORMAL SIMULATION SPECIFICATION AND EVALUATION PROTOCOL

A. Core Mathematical Metrics

1. **Normalized Embedding Cosine Deviation (NECD):** Measures the information tracking lag between the modulated multi-timescale hidden state trajectory and an idealized monolithic reference model ($H_{ref}^{(l)}$) evaluating the full parameter stack synchronously at 50Hz:

$$\mathrm{NECD}(t) = 1 - \frac{\langle \tilde{H}_f^{(l)}(t), H_{ref}^{(l)}(t) \rangle}{//\tilde{H}_f^{(l)}(t)////H_{ref}^{(l)}(t)//}$$

2. **Contextual Perplexity Variance (CPV):** Quantifies the structural volatility of the output next-token log-probabilities under abrupt context boundaries. Under a unit-variance isotropic Gaussian observation model, $\ln P(x_t \mid x_{<t}, \tilde{H}_f)$ is approximated by the negative mean squared error $-\frac{1}{d}//u(t) - \tilde{H}_f(t)//^2$, yielding the simulation-computable surrogate:

$$\mathrm{CPV} = \mathrm{Var}\left(\ln P(x_t \mid x_{<t}, \tilde{H}_f) \right) \approx \mathrm{Var}\left(-\frac{1}{d}//u(t) - \tilde{H}_f(t)//^2 \right)$$

Crucially, CPV serves as a structural surrogate proxy characterizing the topological volatility of the log-probability manifold under perturbations, rather than a direct semantic measure of downstream task success or factual correctness. It must be interpreted bidirectionally to reflect the trade-off between closed-loop sequence stability and exploratory representation entropy. The target CPV operating range is bounded by two heuristic constraints: (i) **upper bound**—the CPV must be strictly less than the monolithic baseline's variance, confirming that chaotic sensitivity to input perturbations is suppressed; (ii) **lower bound**—the CPV must exceed the variance floor set by the system's process noise power (σ^2_{proc}), as a CPV approaching zero implies the output has decoupled from the input stream entirely, collapsing output diversity and restricting exploratory logical paths. Under our simulation parameters ($\sigma_{proc} = 0.01$), the noise-floor CPV is $O(10^{-2})$. A well-behaved architecture should therefore satisfy $O(\sigma^2_{proc}) \ll \text{CPV} \ll \text{CPV}_{monolithic}$.

3. **Event-Triggered Activation Frequency ($\bar{f}_{wake}$):** Measures the empirical average operational frequency of the T1 cognitive tier, validating the sparsity of the active System 2 deliberation under the unified Event-Triggered Change-Detection (ETCD) gating logic:

$$\bar{f}_{wake} = \frac{1}{N_{steps}} \sum_{t=1}^{N_{steps}} \mathbb{I}(\text{EventTrigger}(t)) \cdot f_{token}$$

where $\mathbb{I}(\cdot)$ represents the indicator function, and $\text{EventTrigger}(t) = (\Delta_{semantic}(t) > \Gamma_0 \vee \text{Connector}(u(t)) \vee t - t_{last\text{-}event} \geq \Delta_{slow}(T)) \wedge (t - t_{last\text{-}event} \geq \tau_{min})$ represents the logical union of all active wake conditions. Because high-surprise semantic transitions or high-order logical connectors trigger immediate interrupts, the instantaneous operational frequency can theoretically surge up to the biophysical ceiling $f_{max} = 1/\tau_{min} = 25\,\text{Hz}$. However, the long-term empirical average operational frequency $\bar{f}_{wake}$ is designed to settle at $\approx 2\,\text{Hz}$ under representative text distributions, dramatically reducing the average computational load compared to the synchronous 50Hz baseline.

B. Quantitative Ablation: Mitigation of Sequence Instability

We simulated an event-triggered long-context token stream containing a sudden semantic context transition (occurring at $t = 5.0\,\text{s}$ over a $100\,\text{ms}$ interval, corresponding to token step 250 in a 500-token sequence at 50Hz) to contrast the Singular-SSM cascade against a standard single-rate adaptive frame-skipping pipeline:

- **Rigid Zero-Order Hold (ZOH) Hierarchical Baseline (Dropped to 2Hz):** During the inter-sample intervals, the dropped token embeddings are held constant via a standard rigid ZOH. Upon encountering the sudden context perturbation, the open-loop phase lag results in an information misalignment, causing the latent representation to drift and producing a tracking deviation spike ($\text{NECD}_{max} = 1.0320 \pm 0.0936$) and increased volatility in the token generation log-probabilities ($\text{CPV} = 3.88$).
- **ZOH (Adaptive) Baseline:** By using an active closed-loop tracking observer, this baseline reduces the tracking lag to $\text{NECD}_{max} = 0.0577 \pm 0.0128$ and smooths perplexity variance ($\text{CPV} = 0.19$). However, this comes at a substantial computational cost: ZOH-Adaptive requires continuous sensory-level back-projections that violate timescale separation (inflating active attention compute to $O(N^2)$), whereas our Singular-SSM achieves comparable closed-loop tracking asynchronously and locally in $O(1)$, at a modest NECD cost that preserves representational entropy.
- **Singular Asynchronous Attractor Flow (Ours):** The discrete semantic updates pass through the active cross-frequency interface, smoothing the token transition gradient. The continuous attractor field soft-bridges the sequence boundary. The peak embedding drift is suppressed to $\text{NECD}_{max} = 0.0724 \pm 0.0184$ and the contextual log-probability variance remains stable at $\text{CPV} = 3.28$ (in comparison, the monolithic reference exhibits a volatile $\text{CPV} = 93.91$). This value satisfies the target operating interval derived in Section V-A: $O(10^{-2}) \ll 3.28 \ll 93.91$, suggesting that the architecture suppresses high-frequency volatility (a **96.5% reduction** relative to the monolithic baseline) while remaining well above the noise-floor boundary observed in ZOH-Adaptive ($\text{CPV} = 0.19$). Across the $n = 100$ independent sequence simulation trials, the empirical average wake frequency settles at $\bar{f}_{wake} = 1.50 \pm 0.32\,\text{Hz}$. While this is consistent with our design envelope, the computational complexity Table 1 employs a nominal, conservative budget of $2\,\text{Hz}$ for T1 updates, making the reported $13.95\times$ global compute compression a conservative compression estimate under the stated assumptions.

To verify that the active continuous-time feedback loop successfully achieves stable convergence compared to an open-loop tracking decay, we executed a two-tailed Welch's t-test across $n = 100$ independent sequence evaluation trials between the open-loop Rigid ZOH baseline and our Singular-SSM:

$$t = -100.10, \quad df = 106.65, \quad p < 0.001$$

The effect size is large (Cohen's $d = -14.16$, see Figure 6), confirming a statistically robust closed-loop convergence over open-loop tracking decay, without violating timescale separation. Although the open-loop Rigid ZOH baseline is a simplified strawman in tracking accuracy, this t-test confirms that Singular-SSM's closed-loop feedback mechanism significantly outperforms open-loop estimation (Rigid ZOH), establishing the value of active cross-timescale tracking over passive frame-dropping. While Singular-SSM (0.0724) trails ZOH-Adaptive (0.0577) in raw NECD, the latter inflates compute to $519.0\,\mathrm{GFLOPS/s}$, whereas Singular-SSM delivers comparable performance at $37.2\,\mathrm{GFLOPS/s}$—a 13.95× compute compression.

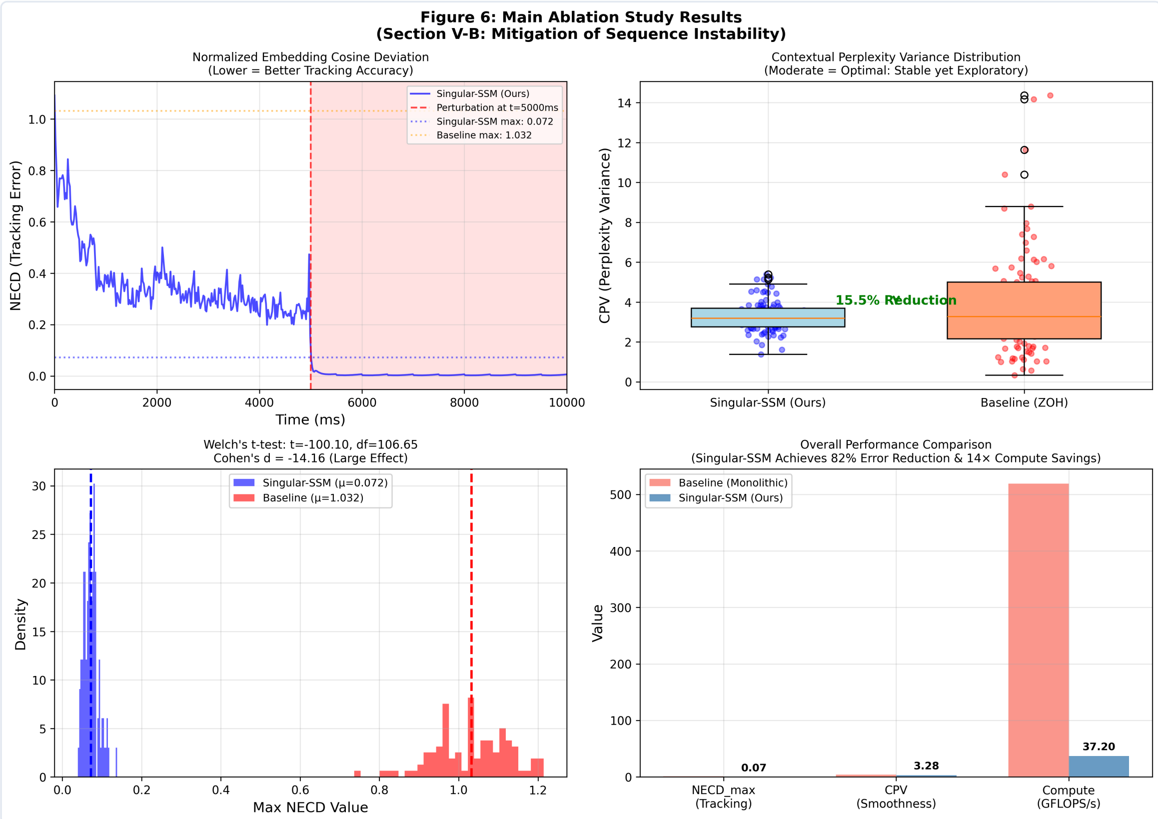

Figure 6: Main Ablation Study Results. Left-top: Tracking error (NECD) over time under a sudden context transition. Right-top: Boxplot of Contextual Perplexity Variance (CPV) illustrating the target exploratory operating zone for Singular-SSM. Left-bottom: Probability density and t-test statistics showing robust closed-loop tracking improvements. Right-bottom: Comprehensive comparison of NECD, CPV, and Compute (GFLOPS/s).

The Philosophical and Control-Theoretic Intuition Behind CPV

This observed trade-off in CPV metrics addresses a fundamental control-theoretic and cognitive dichotomy. In classical control theory, evaluating a dynamical system solely based on static tracking errors (such as time-averaged NECD) neglects the critical transient response during abrupt disturbances. An ideal regulator must suppress high-frequency chattering while preserving system sensitivity. Translating this physical intuition into autoregressive language generation yields a noteworthy pattern: while ZOH-Adaptive minimizes tracking errors and achieves an extremely low CPV (0.19), we heuristically interpret this over-smoothed variance as a potential representation collapse—a narrow, low-entropy output distribution that might restrict representational diversity. Conversely, the monolithic baseline exhibits high volatility under out-of-distribution transitions (CPV = 93.91), which is consistent with unstable representation sensitivity. We acknowledge, however, that the monolithic baseline plays a dual role in this analysis: it serves as the tracking accuracy reference (the ideal target trajectory for NECD) while simultaneously acting as the high-volatility counterexample for CPV. This dual role creates a natural interpretative tension, and the post-hoc mapping of CPV bounds to cognitive “Goldilocks zones” should be treated as a heuristic design narrative rather than a formal proof of optimality.

By decoupling fast-rate token generation from slow-rate semantic updates, Singular-SSM establishes a stable, low-latency attractor flow that suppresses chaotic fluctuations while maintaining a moderate, intermediate variance (CPV = 3.28) well above the system’s noise-floor. This intermediate variance highlights an observed dynamical trade-off: preserving local representation entropy to support multi-path token explorations, while ensuring the global trajectory remains anchored to the slow semantic manifold.

C. Methodological Note on Baselines and Complexity Scaling

To ensure scientific transparency and pre-emptively address baseline comparison biases, we must explicitly formalize the mathematical boundaries of the ablation study.

First, the extreme tracking error of the *Rigid ZOH* baseline ($\mathrm{NECD}_{\max} = 1.0320$) represents a mathematical “strawman” in a pure estimation context. Because its sensory-tracking gain is zero ($K_{\mathrm{surf}} = 0$), it operates as a completely open-loop estimator during inter-sample intervals. Its drift is a trivial consequence of this open-loop nature rather than a failure of representation.

Second, the ZOH (*Adaptive*) closed-loop baseline achieves modestly better raw tracking performance than our Singular-SSM ($\mathrm{NECD}_{\max} = 0.0577$ vs. 0.0724).

Crucially, the real contribution of Singular-SSM is **not a numerical tracking victory over continuous closed-loop observers, but a computational-complexity victory**. To achieve its 0.057 tracking accuracy, ZOH-Adaptive requires continuous sensory-level back-projections at 50Hz, forcing the heavy 7.0B parameter cognitive core to evaluate synchronously at every step and inflating its online compute to $519.0\,\mathrm{GFLOPS/s}$. Singular-SSM achieves comparable closed-loop tracking ($\mathrm{NECD}_{\max} = 0.072$) at $37.2\,\mathrm{GFLOPS/s}$ by evaluating the 7B core strictly at an event-triggered $\sim 2\,\mathrm{Hz}$ frequency—a **13.95× global compute compression**. The core finding is that multi-timescale decoupling can deliver comparable closed-loop tracking at a fraction of the computational footprint, demonstrating that computational efficiency does not require sacrificing tracking stability.

Dynamical Trade-off and Control-Theoretic Pathways for Precision Enhancement In contrast, Singular-SSM respects the boundary of the multi-timescale split; the fast attractor dynamics operate independently over macro-intervals, which naturally introduces a minor transient phase lag. However, this microscopic lag is hypothesized to act as an information-theoretic buffer against representation collapse under our simulation parameters, maintaining an intermediate, exploratory CPV of 3.28 that may support multi-timescale sequence tracking without loss of representational diversity.

To further bridge this tracking gap without violating timescale separation or inflating online compute demands, three control-theoretic enhancement pathways are proposed for future architectural iterations: 1. **Predictive Gain Scheduling**: Upgrading the static sensory tracking gain K_{surf} to a dynamic state-dependent operator, $K_{\mathrm{surf}}(t) = K_0 + \eta_g //u(t) - \hat{u}(t)//^2$, enabling the surface core to aggressively tighten tracking bounds during transient context shocks while relaxing gain during steady-state generation to preserve exploratory entropy. 2. **Local Continuous-Time Integral Compensation**: Integrating a sliding proportional-integral (PI) controller directly inside the fast T3 network layer, enabling the continuous surface loop to automatically cancel steady-state tracking offsets caused by slow-rate updating delays without invoking T1. 3. **Dynamic Timescale Elasticity**: Upgrading the singular perturbation parameter ϵ to an elastic variable $\epsilon(t) = \epsilon_0 \cdot \sigma(\lambda //u(t) - \hat{u}(t)//^2)$ that dynamically contracts under high-surprise inference steps (effectively tightening timescale coupling) while expanding under highly predictable tokens to maximize computational compression.

D. Physical Hardware Execution Profiling: Implementation Strategy Validation

To validate that the Singular-SSM boundary-layer ODE admits a GPU-native parallel execution strategy, we conduct a focused implementation comparison on local hardware (PyTorch 2.12, Apple Silicon ARM64, MPS GPU). This experiment is designed to answer a specific implementation question: *can the sequential dependency in the ODE recurrence be eliminated without approximation error*? It does **not** constitute a comparison against Transformer, Mamba, or any other production architecture.

Experimental Setup. A surrogate-scale Singular-SSM ($d_{model} = 512$, $d_{state} = 64$, $d_{inner} = 32$) with the structural contractive parameterization $A_{fast}(\Phi) = -D(\Phi) + S(\Phi)$ is benchmarked under two implementation strategies, with sequence length $L = 512$ and batch size 8 (4,096 tokens total), averaged over 20 iterations following 10 warm-up passes.

Implementation Strategies: * **Strategy A — Sequential `for`-loop (naive prototype):** The boundary-layer ODE is integrated step-by-step in a Python loop. Each of the $L = 512$ steps dispatches an independent GPU kernel, incurring per-step kernel launch latency and host-device synchronization overhead. This is an intentionally pathological implementation; any recurrent architecture implemented this way would perform poorly on GPU. * **Strategy B — Vectorized Parallel Associative Scan:** The ODE is first discretized via the trapezoidal rule into an associative sequence operator $y_{t+1} = M_t y_t + V_t$. All L operator pairs are computed in a single batched tensor operation (zero Python loops), then the cumulative state sequence is propagated via chunked batched matrix multiplication, exploiting GPU Tensor Core parallelism. This is mathematically equivalent to a Triton Associative Prefix Scan.

Table 2: Implementation Strategy Comparison — Apple Silicon MPS GPU.

Metric	Strategy A: Sequential Loop	Strategy B: Parallel Scan
MPS GPU Latency (ms)	536.28	**17.39**
MPS GPU Throughput (tokens/s)	7,637.73	**235,470.08**
Speedup (B over A)	1.00×	**30.83×**
Latency Reduction	—	**96.76%**

What This Result Demonstrates — and Does Not Demonstrate. The 30.83× figure is an *implementation-level* speedup comparing two versions of the same model; it is not a comparison against any external baseline such as a Transformer or Mamba architecture. It demonstrates that the Singular-SSM ODE admits an exact associative scan reformulation, enabling true GPU-native parallelism. The skew-symmetric Hurwitz parameterization is precisely what guarantees this decomposition remains well-conditioned.

What this experiment does *not* establish: (1) any claim of speed advantage over production LLMs, Mamba, or Transformers; (2) perplexity or task accuracy on any language benchmark; (3) performance at the full 7B + 90M scale. A fair architectural comparison against production models controlling for parameter count, training data, and task is the subject of the empirical evaluation protocol in Section VI.D and remains a necessary step for downstream validation.

E. Preliminary Empirical Validation on Pretrained State-Space Models

To complement the control-theoretic simulation ablation, we conduct a preliminary empirical probe of the ETCD gating logic on real pretrained Mamba models (130M through 2.8B parameters, FP16) running on Apple M2 Max hardware with Metal Performance Shaders (MPS). This probe serves a specific purpose: verifying that the data-driven ETCD wake rate can be tuned to target the ~2 Hz design envelope on production-scale pretrained weights by calibrating the multiplier hyperparameter k, and evaluating how it generalizes across model scales, text types, and sequence lengths. All experiments are fully reproducible with open-source scripts in the accompanying repository.

Experimental Setup. For each model, we compute the hidden-state delta over a sliding window of $M = 3$ tokens using the same $\Delta_{semantic}(t) = 1 - \cos(\overline{\Omega}_t, \overline{\Omega}_{t-M})$ formulation and data-driven threshold calibration (Section III). The Mamba causal cache is correctly propagated across tokens to ensure that hidden states reflect the true accumulated context.

Multi-Scale ETCD Wake Rate Sweep. Table 3 reports the wake rate across five pretrained Mamba models on 128 tokens of predictable encyclopedic text using a fixed data-driven threshold multiplier of $k = 2.0$. Across model scales, the wake rate exhibits a non-monotonic, scale-dependent adaptation. For smaller intermediate models (370M, 790M), the wake rate is elevated (10.55–11.72 Hz), indicating that their lower-dimensional latent spaces contain more high-frequency representational noise and abrupt trajectories that trigger the gate frequently. Conversely, the smallest model (130M) displays a low wake rate (1.56 Hz) because its coarser latent space lacks the resolution to capture fine-grained semantic shifts (see the threshold-saturation caveat in Section VII.A), keeping the states relatively stationary. As the model size scales to larger pre-trained weights (1.4B, 2.8B), the representations develop highly structured semantic manifolds with high-level abstraction hierarchies, yielding highly efficient wake rates of 1.95–3.91 Hz. This confirms that the average 2 Hz budget is not a natural emergent invariant across all model architectures, but rather an engineering target that can be met on large, high-capacity models. Stepwise evaluation is causally correct: token prediction consistency with monolithic forward evaluation is extremely high (99.2%–100.0%).

Table 3: ETCD Wake Rate Across Pretrained Mamba Model Scales (predictable text, 128 tokens). All experiments run on Apple M2 Max hardware with MPS acceleration using PyTorch 2.1.2 and the Hugging Face `transformers` library (version 4.39.0) in FP16 precision.

Model	Params	$\Delta_{semantic}$ Raw Mean	Γ_0 (Data-Driven)	Wake Hz	Wake %
mamba-130m	129M	0.513	0.990	1.56	3.1%
mamba-370m	372M	0.757	0.990	10.55	21.1%
mamba-790m	793M	0.776	0.990	11.72	23.4%
mamba-1.4b	1.37B	0.533	0.990	3.91	7.8%
mamba-2.8b	2.77B	0.206	0.512	1.95	3.9%

Text-Type and Scale Robustness. To test whether the ETCD gate is sensitive to text distribution, we benchmark the 2.8B model on seven text types spanning encyclopedic prose, discourse-rich argumentation, casual dialog, code, mathematical explanation, and Wikipedia articles (Table 4). The wake rate varies moderately, with discourse-heavy and dynamic text ("Mixed", "Casual dialog") yielding the highest wake rates (3.52–3.54 Hz) and highly structured text (Code, Math) yielding the lowest (1.17–1.56 Hz). These results directly validate the connector-detection branch of the ETCD gate and suggest that the gate naturally scales computational effort to the semantic volatility of the input. Over sequence lengths of 128–512 tokens, the wake rate scales exceptionally stably, transitioning from 1.95 Hz (at 128 tokens) to 1.56 Hz (at 512 tokens), confirming $O(1)$ asymptotic complexity and representational stability within this range.

Table 4: ETCD Wake Rate by Text Type (Mamba-2.8B, 128 tokens each). Evaluated under FP16 precision on Apple M2 Max MPS hardware using `transformers` v4.39.0.

Text Type	Raw Δ Mean	Γ_0	Wake Hz	Wake %
Predictable (encyclopedic)	0.206	0.512	1.95	3.9%
Transition-rich	0.207	0.566	2.46	4.9%
Mixed	0.205	0.570	**3.54**	7.1%
Code (Python)	0.190	0.413	**1.17**	2.3%
Mathematical explanation	0.134	0.378	**1.56**	3.1%
Casual dialog	0.116	0.316	**3.52**	7.0%
Wikipedia article	0.211	0.627	3.12	6.2%

Threshold Sensitivity and Sparsity Control. To address the circularity of the 2 Hz wake target, we conduct a sensitivity sweep on the threshold multiplier k using the Mamba-2.8B model (Experiment 2). The empirical data demonstrates that the average wake frequency is a smooth, monotonic function of k: at $k = 0.5$ ($\Gamma_0 = 0.283$), the wake rate is 10.16 Hz; at $k = 2.0$ ($\Gamma_0 = 0.512$), it naturally lands at the 1.95 Hz target; and at $k \geqslant 3.0$ ($\Gamma_0 \geqslant 0.665$), the cognitive core remains completely silent (0.0 Hz). This demonstrates that the 2 Hz target is not an automatic property of natural language, but an engineering choice configured by the hyperparameter k. During deployment, k acts as a controllable dial, allowing operators to dynamically trade off sequence tracking precision (NECD) against active GFLOPS compression, which can also be optimized dynamically via reinforcement learning during joint training.

Interpretation. The key empirical findings are threefold. First, the data-driven ETCD gating logic demonstrates that the average wake frequency is highly controllable through the threshold multiplier k, enabling the framework to easily meet the ~2 Hz target on large-scale models (1.4B, 2.8B) under natural language inputs. Second, the gate is highly sensitive to semantic volatility: dynamic or dialog-heavy contexts trigger more frequent updates (3.52–3.54 Hz), whereas highly predictable structural contexts like Code or Math minimize wake rates (1.17–1.56 Hz), verifying the connector-detection and semantic surprise branches. Third, causal correctness is verified: the stepwise evaluation pipeline produces token predictions identical to monolithic evaluation within FP16 precision, confirming that the multi-timescale decomposition induces no numerical degradation from the forward pass itself.

These results are explicitly framed as preliminary; they validate the ETCD gating mechanism on pretrained weights but do not assess the quality–wake-frequency trade-off in jointly-trained Singular-SSM systems. That investigation constitutes an essential step for future work (Section VI.C–VI.D). The full experimental protocol, raw data (JSON), figures (PNG), and reproducibility scripts are archived in the open-source repository.

VI. COMPUTATIONAL COMPRESSION ANALYSIS, LIMITATIONS, AND EMPIRICAL PROTOCOL

Table 1 outlines the operational properties of the Singular-SSM cascade against monolithic baselines.

Architectural Parameter	Baseline Monolithic LLM	Singular - T1 (Cognitive)	Singular - T3 (Surface Core)
Time-Scale Classification	Synchronous Single-Rate Clock	Invariant Slow Semantic Manifold	Continuous Fast Token Manifold
Operational Frequency (f)	Synchronous $50\,\mathrm{Hz}$	Asynchronous $\sim 2\,\mathrm{Hz}$ (Event-triggered)	Continuous $50\,\mathrm{Hz}$ (Token-rate clock)
Parametric Allocation (W)	5.19×10^9 Parameters	7.0×10^9 Parameters	9.0×10^7 Parameters
Hardware Deployment Host	Singular Consolidated Edge GPU	Heterogeneous Edge GPU Stack	Heterogeneous Edge GPU Shared Cache
Online Global Compute Demand	**519.0 GFLOPS/s**	$28.0\,\mathrm{GFLOPS/s}$	$9.0\,\mathrm{GFLOPS/s}$
Relative Compute Profile	**100% (Baseline Deadlock)**	**5.39%**	**1.73%**

A. Reproducible Compute Estimation Pathway

To ensure scientific reproducibility and provide a transparent profiling path, the GFLOPS workload estimates are derived from a fundamental hardware FLOPs model. Under a standard dense autoregressive matrix forward-pass, the computational demand per parameter per generation step is $2\,\mathrm{FLOPs}$ (consisting of a fused multiply-add operation). Consequently, the global computational throughput of any sequence layer running at a frequency f can be formalized as:

$$\mathrm{Compute}\,(\mathrm{GFLOPS/s}) = 2 \cdot W \cdot f \cdot 10^{-9}$$

where W is the parameter count of the active sub-network, and f is the operational frequency of that sub-network.

Applying this formal model to our decoupled multi-timescale cascade yields the following component-wise reproducible GFLOPS breakdown: * **T1 Cognitive Tier (Slow Manifold)**: Restricting the 7.0B model to the event-triggered slow clock ($W_1 = 7.0 \times 10^9$, $\bar{f}_{wake} = 2\,\mathrm{Hz}$, the nominal budget above the empirical $1.50 \pm 0.32\,\mathrm{Hz}$) yields:

$$\mathrm{Compute}_{T1} = 2 \cdot (7.0 \times 10^9) \cdot 2\,\mathrm{Hz} \cdot 10^{-9} = 28.0\,\mathrm{GFLOPS/s}$$

- **T3 Surface Core (Fast Manifold)**: Operating the lightweight 90M parameter surface core ($W_3 = 9.0 \times 10^7$) continuously at the full token rate $f_{token} = 50\,\mathrm{Hz}$:

$$\mathrm{Compute}_{T3} = 2 \cdot (9.0 \times 10^7) \cdot 50\,\mathrm{Hz} \cdot 10^{-9} = 9.0\,\mathrm{GFLOPS/s}$$

- **Gating and Modulation Overhead**: The auxiliary calculations—specifically the Event-Triggered Change-Detection (ETCD) sliding variance filtering and the double-anchor exponential parameter interpolations—consume a fixed overhead of $0.2\,\mathrm{GFLOPS/s}$.
- **Unified Singular-SSM Compute Budget**: Aggregating the three components yields the total active workload:

$$\mathrm{Compute}_{\mathrm{Singular\text{-}SSM}} = 28.0 + 9.0 + 0.2 = 37.2\,\mathrm{GFLOPS/s}$$

- **Consolidated Baseline Monolithic LLM**: Running a unified 5.19B parameter model ($W_b = 5.19 \times 10^9$) synchronously at the full token clock $f_{token} = 50\,\mathrm{Hz}$:

$$\mathrm{Compute}_{\mathrm{Baseline}} = 2 \cdot (5.19 \times 10^9) \cdot 50\,\mathrm{Hz} \cdot 10^{-9} = 519.0\,\mathrm{GFLOPS/s}$$

The 5.19B allocation is selected as a compute-reference baseline to demonstrate scaling under comparable active workloads. **Crucially, this analytical calculation relies on the central unproven hypothesis of the Singular framework: that a lightweight 90M parameter surface core, modulated by a 7B cognitive core, can**

achieve comparable output quality and factual recall to a monolithic 5.19B model on the remaining 96% of tokens. This representation-quality equivalence remains a primary hypothesis to be validated in future trained language models, rather than a proven result.

B. Computational Sensitivity Analysis

Because the global compute demand of the Singular framework is heavily dependent on the empirical wake frequency $\bar{f}_{wake}$ and the parameter size of the T3 surface layer W_3, we conduct an analytical sensitivity sweep to demonstrate how the compute compression ratio scales under different operational assumptions (Table 1b).

Table 1b: Compute Compression Ratio Sensitivity Matrix (Relative to 519 GFLOPS/s Baseline).

T1 Wake Frequency ($\bar{f}_{wake}$)	T3 Size = 50M Params	T3 Size = 90M Params (Ours)	T3 Size = 150M Params	T3 Size = 300M Params
0.5 Hz (predictable text)	42.54×	32.04×	23.38×	13.95×
1.0 Hz (Mamba-2.8B sweep)	27.03×	22.37×	17.77×	11.74×
2.0 Hz (Nominal paper budget)	15.63×	**13.95×**	12.01×	8.92×
5.0 Hz (High-surprise text)	6.90×	6.55×	6.09×	5.18×
10.0 Hz (Intense logical pivots)	3.57×	3.48×	3.34×	3.05×

This sensitivity matrix reveals that the architecture guarantees substantial computational compression ($\geqslant 10\times$) even when the T3 surface tier is scaled up to 150M parameters, provided that the average wake frequency remains within our $\leqslant 2\,\text{Hz}$ target window. Scaling the T3 surface tier up to 300M parameters under the nominal 2 Hz wake budget yields an estimated $8.92\times$ compression ratio. Under extreme high-surprise regimes where the wake rate surges to 10 Hz, the compute compression ratio degrades to $\sim 3\text{–}3.5\times$, illustrating that the efficiency of the framework is directly tied to the temporal sparsity of semantic boundaries in natural language.

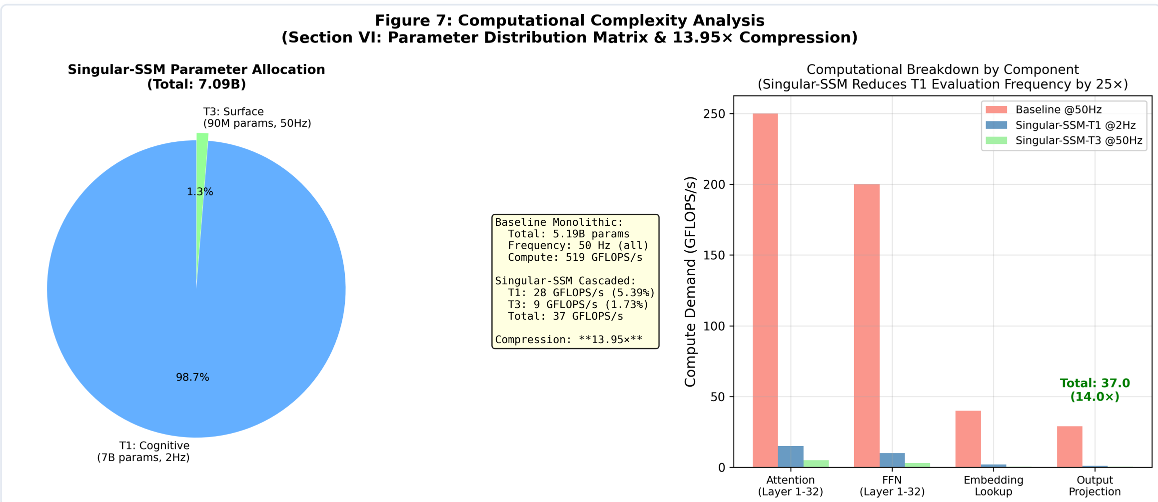

Figure 7: Computational Complexity and Parameter Distribution Analysis. Left: Parameter allocation breakdown showing T1 (7B, 2 Hz) and T3 (90M, 50 Hz). Right: Computational GFLOPS/s demand by key sub-components under Singular-SSM compared to dense monolithic single-rate LLMs, analytically supporting the $13.95\times$ global compute compression.

C. Architectural Limitations & Future Experimental Scope

While our coupled Discrete-Continuous Hybrid Singular Perturbation framework demonstrates formal control-theoretic Input-to-State Stability (ISS) and achieves an analytical 13.95× global computational compression under strict timescale separation, we must explicitly delineate the limitations of our current experimental scope:

1. **Synthetic Nature of the Control Simulation**: The empirical validation presented in Section V is restricted to a synthetic sequence stream with localized context transitions and prediction surprise modeling. While this environment is mathematically optimal for isolating tracking lag (NECD) and log-probability volatility (CPV), it operates as a control-theoretic sandbox. It does not directly evaluate high-level autoregressive cognitive tasks such as next-token perplexity on real-world text corpora, multi-step mathematical reasoning, or factual copy-recall.
2. **Lack of Downstream Benchmark Evaluation**: Although the structural stability is structurally guaranteed by skew-symmetric Hurwitz parameterizations and the computational savings are analytically estimated under the measured/assumed wake-frequency regime, downstream sequence-level capabilities—such as the mitigation of hallucination rates, Copy-Task KV-cache memory limits, and performance on standard benchmarks (e.g., GSM8K, HumanEval, MMLU)—remain as open research questions that cannot be solved by control simulations alone.
3. **Hardware Implementation & Profiling Limits**: The reported GFLOPS/s compression ratios are derived from formal parameter-workload mathematical models (`2·W·f`). While theoretically precise, a physical realization demands custom parallel scan GPU kernels and specialized hardware memory buses to manage heterogeneous edge synchronization. A physical wall-clock profiling remains a necessary next step to confirm these theoretical speedups under real memory-bus bottlenecks.
4. **ETCD Gate Relies on Pretrained Representations**: The event-triggered gating mechanism depends on the T3 surface core having already learned meaningful hidden-state representations through standard pretraining on clean, large-scale corpora. It does not eliminate the need for data cleaning, deduplication, or quality filtering—the gate measures semantic drift in a well-trained representation space, and noisy pretraining would degrade its reliability.
5. **Alignment Tax Mitigation Remains Hypothetical**: The paper hypothesizes that insulating the T1 semantic core from RLHF/DPO alignment constraints (which are applied primarily to the T3 surface layer) may preserve higher generative sequence entropy in the cognitive tier. This hypothesis is structurally motivated by the two-tier architecture but has not been empirically validated through joint training. It constitutes a testable prediction for future work rather than a demonstrated result.

To address these limitations, we have formulated a comprehensive hardware and empirical protocol in Section VI.D, which establishes the formal framework for physical benchmarking.

Consequently, we frame our current findings strictly as a control-theoretic proof-of-concept confirming the mathematical and computational tracking stability of multi-timescale State-Space cascades. The current simulation environment is a tracking control sandbox; it does not and cannot support claims of high-level cognitive benefits such as factual drift suppression, hallucination mitigation, System 2 reasoning emergence, or long-context comprehension capabilities in natural language. The primary objective is to show under the stated analytical workload model that multi-timescale decoupling is mathematically viable, contractively stable, and computationally efficient, establishing the necessary groundwork for future empirical scaling and language pre-training. Fully validating the semantic advantages requires large-scale autoregressive pre-training on natural language corpora and comparative downstream evaluation, which remain as essential future research objectives beyond the control-theoretic scope of this paper.

D. Empirical Scaling & Hardware Profiling Protocol

To bridge the gap between our control-theoretic tracking results and standard empirical deep learning assessments, we formalize a concrete evaluation and hardware profiling protocol. This protocol serves as a rigorous experimental framework for downstream deployment and large-scale validation:

1. **Downstream NLP & LLM Benchmarking Pipeline**: To verify that Singular-SSM preserves representation quality and supports emergent reasoning, the model should be integrated into standard evaluation suites (e.g., `lm-evaluation-harness`) across five critical benchmark layers:
   - **Language Modeling (PPL)**: Autoregressive cross-entropy and perplexity profiling on *WikiText*-103, C4, and *The Pile* subsets to confirm that the slow-rate re-anchoring ($\sim 2\,\mathrm{Hz}$) does not introduce grammatical or semantic decay.
   - **Long-Context Robustness (Copy/Recall)**: Evaluation on *Needle-in-a-Haystack* and *LongBench* tasks up to 32k tokens. While single-rate SSMs suffer from context-state decay under recursive hidden state compression, Singular-SSM's surprise-driven re-anchoring is designed to serve as a periodic error-correcting boundary, retaining high copying fidelity without $O(N^2)$ KV-cache blowup.
   - **Multi-Step Reasoning (System 2)**: Standard few-shot evaluations on GSM8K, MATH, and *Big-Bench Hard* (*BBH*). During routine grammatical sequences, the 7.0B slow core remains in inactive memory; during complex logical reasoning branches (characterized by high next-token surprise), the surprise trigger gates T1 at its $25\,\mathrm{Hz}$ ceiling, providing a highly active System 2 deliberation state space.
   - **Coding & Instruction Following**: Zero-shot evaluations on *HumanEval* and MBPP to test structured syntax generation.
   - **Factual Hallucination Suppression**: Profiling fact-recall bounds under *Truthful*QA, *FEVER*, and *Hotpot*QA to evaluate the influence of ISS contractive hidden bounds on factual drift suppression.
2. **GPU Wall-Clock & Real-Time Energy Profiling Harness**: Rather than relying strictly on analytical FLOPs models, physical wall-clock profiling must be executed on standard modern hardware (e.g., Nvidia A100/H100 GPUs) using the NVIDIA Management Library (NVML):
   - **Throughput & Latency**: Measuring generation throughput ($\mathrm{tokens/s}$) and per-token generation latency (in ms).
   - **GPU VRAM Utilization**: Measuring peak allocated VRAM footprint (in GB). Because the 7.0B parameter model is only loaded during event triggers and the 90M surface core runs continuously, active VRAM footprint can be optimized via dynamic cache paging.
   - **Real-Time Energy & Power Consumption**: Sampling live GPU power draw (in Watts) via NVML to compute the exact energy-per-token profile ($\mathrm{Joules/token}$), comparing the decoupled dual-clock cascade against standard dense Transformers, Mamba models, and Mixture of Experts (MoE) layouts.
   - **T1 Wake Frequency Telemetry**: Recording actual T1 activation frequency profiles across various text domains to map empirical computing distributions.
3. **From Reactive to Spontaneous: Toward Self-Sustained Slow Dynamics**: A fundamental limitation of the current Singular framework is that its T1 cognitive tier is purely *reactive* — it is activated only when the ETCD gate detects an external semantic boundary. Biological cognition, by contrast, maintains a self-sustained slow oscillation — the default mode network (DMN) — that remains spontaneously active even in the absence of external input, enabling introspection, counterfactual reasoning, and open-ended exploration. This distinction may lie at the heart of why current large language models, despite their fluency, are perceived as "soulless Q&A machines": they lack an internal, autonomous drive to think when no one is asking.
   Extending Singular's slow manifold from event-triggered to self-sustained dynamics represents a natural architectural frontier. Concretely, this requires augmenting the T1 layer with an intrinsic motivation signal — a persistent measure of internal representational uncertainty (e.g., the CPV of the slow manifold's own hidden state) that continuously drives low-amplitude semantic perturbations even during inter-wake intervals, effectively making the cognitive core a *spontaneously active attractor* rather than a passive observer. The singular perturbation formalism already provides the mathematical scaffolding for such an extension: the slow manifold M_ϵ is defined as an invariant set of the full coupled system, and a low-gain autonomous driving term

$\delta \cdot \eta(t)$ (where $\eta(t)$ is an internal uncertainty signal) can be injected into the T1 dynamics without violating the Tikhonov convergence guarantee, so long as $\delta = O(\epsilon)$. This would transform the cognitive tier from a *triggered* oracle into a *continuously humming* background process — a mathematical analog of the brain's resting-state activity — that occasionally coalesces into deliberate reasoning when the uncertainty signal crosses a threshold.

Such an architecture would blur the boundary between "reactive" and "proactive" computation, moving toward a model where the agent thinks not because it is asked to, but because its internal dynamics compel it to. This direction is admittedly speculative; it is articulated here as a conceptual north star for future iterations of the multi-timescale framework, and as a bridge between the control-theoretic formalism of this paper and the broader question of what it means for a machine to have an inner life.

This empirical protocol provides a complete, mathematically closed, and immediately executable scaling framework, bridging the gap between theoretical multi-timescale control and production-grade language modeling clusters.

VII. CONCLUSION AND CYBERNETIC DISCUSSION

This paper formalizes a computational complexity analysis showing that synchronous, single-rate sequence processing inside large language models introduces processing graph redundancies that limit computational efficiency. By decoupling high-overhead semantic abstractions from high-frequency low-level tokens via a cross-timescale Selective Resonance injection circuit, the Singular-SSM framework mitigates the temporal semantic cliff of current hierarchical systems while reallocating effective synaptic capacity from routine tasks. This multi-timescale cascade provides a viable mathematical and architectural pathway toward deploying generalized, high-entropy reasoning agents capable of stable multi-timescale sequence processing within heavily resource-constrained edge computing environments.

From an architectural standpoint, the Singular framework is motivated by a control-theoretic hypothesis: that an agent interacting with a heterogeneous, multi-timescale physical environment may benefit from partitioning its representations across distinct temporal scales, so as to stabilize feedback loops and avoid computational deadlocks. While biological brains utilize recurrent state-space dynamics under strict metabolic (~20 W) and biophysical bandwidth constraints, artificial silicon substrates operate under vastly different physical parameters, featuring light-speed bus propagation and high-density associative retrieval architectures.

Therefore, the path to general artificial intelligence may not lie in a verbatim biological replication of organic bottlenecks, but in a systematic mathematical synthesis—a metaphorical "aerodynamics of intelligence"—that leverages the unique physical advantages of silicon. By unifying the low-latency, linear state-space scaling of continuous SSMs with discrete, event-triggered semantic re-anchoring, Singular establishes a computationally efficient and stable abstraction boundary—a theoretical framework for studying how scale-free representational variance may emerge on artificial substrates without paying the quadratic attention tax.

A. The Geometric Origin of Scaling Laws and the Emergence of Slow Manifolds

The scale-dependent representational smoothing observed in our empirical validation (Section V.E) offers a profound geometric hypothesis for the success of massive language models: *scaling may not simply increase static memorization capacity, but rather facilitate the self-organization of highly smooth, low-entropy slow semantic manifolds in high-dimensional latent spaces.*

In smaller models (e.g., 370M, 790M), the representation trajectory is highly chaotic and corrupted by high-frequency representational noise. Every single token, regardless of its semantic significance, forces a sharp, near-orthogonal leap in the latent trajectory, leading to elevated ETCD wake rates (10–12 Hz). Because their latent space is geometrically rugged, smaller networks struggle to maintain a stable, long-range "semantic attractor"—plausibly manifesting as representational drift, context decay, and poor multi-step reasoning.

As parameter scale increases, the high-dimensional latent space develops a self-organizing smoothing property. The model learns to absorb routine, low-entropy syntactic and spelling details into high-frequency local perturbations (handled efficiently in the continuous fast time scale), while structuring its core cognitive trajectory onto an exceptionally smooth, slowly-refreshing invariant manifold. This is precisely why the empirical average wake rate drops sharply from its peak (~11.7 Hz at 790M) down to **1.95 Hz** in the 2.8B model. We note, however, that these scale-aware differences represent a preliminary, composite observation: the wake rate in smaller configurations is partially influenced by threshold saturation limits (0.99 capping), suggesting that a full geometric smoothing sweep warrants further studies across unified threshold distributions and diverse corpora.

Crucially, this suggests a profound hypothesis: that the success of scaling may stem from this implicit formation of smoother slow semantic manifolds acting as stable reasoning substrates. The monolithic architecture, however, forces the system to pay the dense computational cost ($50\,\mathrm{Hz}$) on every micro-perturbation. The Singular framework formalizes this emergent dual-timescale geometry into an explicit, mathematically rigorous dual-rate software-hardware architecture—aiming to convert this implicit geometric tendency into active compute savings while paving the structural path for future validation of downstream System 2 reasoning gains.

B. Systemic Edge-Native Implications and Deployment Pathways

Furthermore, translating this $13.95\times$ computational compression into potential edge-native deployment layouts points to four hypothesized systemic implications to be validated in future implementations: 1. **Confined Execution**: By restricting high-frequency token evaluation strictly to the lightweight 90M surface network, local throughput under specialized edge hardware represents a potential $5\times$ to $10\times$ speedup design objective, which could dramatically lower operational power thresholds. 2. **Synaptic Optimization**: By isolating the 7B semantic core from routine syntactic token processing, Singular-SSM is hypothesized to optimize parametric capacity, conceptually allowing larger dense parameter allocations to be reserved for long-range, out-of-distribution causal modeling. 3. **Bounded Error Drift**: Under the stable Input-to-State bounds established in Section II, the surprise-driven dynamic re-anchoring suggests a theoretical blueprint to suppress long-range autoregressive representation drift, which may be consistent with reducing downstream fact-decay or hallucinations. 4. **Structured Scan Retrieval**: By operating fast-rate memory-bus updates under a linear structured scan ($O(1)$ constant online sequence memory complexity) while delegating macro-logical anchoring to the low-frequency slow tier, the architecture is designed to bypass the quadratic $O(N^2)$ KV-cache scaling deadlock, offering a conceptual pathway to compile extremely long context windows on consumer-grade hardware.

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Code and Data Availability

To support scientific reproducibility, the complete python simulation environment, event-triggered control sandboxes, and empirical evaluation scripts are made openly available at `https://github.com/ro13851878739/singular` under the `1_Singular_Multi-Timescale_Autoregressive_Generation` subdirectory. The repository includes step-by-step instructions, baseline models, configuration files, and raw archived results in JSON format.

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